

# Groups and Lie Algebras

## Solutions to Exercise Sheet 4

### Exercise 1: Projection operators for non-trivial irreducible representations

Let  $(V, \rho)$  be a finite-dimensional representation of a finite group  $G$ . Let  $(V_i, \rho_i)$  be an irreducible representation of  $G$ , and let

$$\chi_i(g) = \text{tr}_{V_i} \rho_i(g)$$

be its character.

The standard projection operator onto the  $V_i$ -isotypic component of  $V$  is

$$P_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g).$$

Equivalently, since finite-dimensional representations of finite groups can be made unitary,

$$\overline{\chi_i(g)} = \chi_i(g^{-1}),$$

so one can also write

$$P_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \chi_i(g^{-1}) \rho(g).$$

If all characters are real-valued, as happens for many examples such as many representations of symmetric groups, this agrees with the formula using  $\chi_i(g)$ .

**Solution.** We want to prove that  $P_i$  is the projection onto the largest subrepresentation of  $V$  which is isomorphic to a direct sum of copies of  $V_i$ . This subspace is called the  *$V_i$ -isotypic component* of  $V$ .

By Maschke's theorem,  $V$  decomposes as a direct sum of irreducible representations:

$$V \cong \bigoplus_j N_j \otimes V_j.$$

Here  $V_j$  ranges over the irreducible representations of  $G$ , and  $N_j$  is a multiplicity space. The group  $G$  acts trivially on  $N_j$  and by the irreducible representation  $\rho_j$  on  $V_j$ :

$$\rho(g)|_{N_j \otimes V_j} = \text{id}_{N_j} \otimes \rho_j(g).$$

The  $V_i$ -isotypic component is therefore

$$V^{(i)} := N_i \otimes V_i.$$

We must show that  $P_i$  acts as the identity on  $V^{(i)}$  and as zero on every other isotypic component  $N_j \otimes V_j$  with  $j \neq i$ .

It suffices to understand the action of  $P_i$  on a single irreducible summand  $V_j$ . On  $V_j$ , the operator becomes

$$P_i|_{V_j} = \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho_j(g).$$

This operator commutes with the action of  $G$ . Indeed, for  $h \in G$ ,

$$\begin{aligned} \rho_j(h) P_i|_{V_j} \rho_j(h)^{-1} &= \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho_j(hgh^{-1}) \\ &= \frac{\dim V_i}{|G|} \sum_{g' \in G} \overline{\chi_i(h^{-1}g'h)} \rho_j(g') \\ &= \frac{\dim V_i}{|G|} \sum_{g' \in G} \overline{\chi_i(g')} \rho_j(g') \\ &= P_i|_{V_j}, \end{aligned}$$

where we used the fact that characters are class functions.

Thus  $P_i|_{V_j}$  is an intertwiner from  $V_j$  to itself. By Schur's lemma, since  $V_j$  is irreducible, this operator must be a scalar multiple of the identity:

$$P_i|_{V_j} = c_{ij} \text{id}_{V_j}.$$

To determine the scalar  $c_{ij}$ , take the trace. We get

$$\begin{aligned} \text{tr}_{V_j}(P_i|_{V_j}) &= \frac{\dim V_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \text{tr}_{V_j}(\rho_j(g)) \\ &= \dim V_i \frac{1}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \chi_j(g). \end{aligned}$$

By the orthonormality of irreducible characters,

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \chi_j(g) = \delta_{ij}.$$

Therefore

$$\text{tr}_{V_j}(P_i|_{V_j}) = \dim V_i \delta_{ij}.$$

On the other hand, if  $P_i|_{V_j} = c_{ij} \text{id}_{V_j}$ , then

$$\text{tr}_{V_j}(P_i|_{V_j}) = c_{ij} \dim V_j.$$

Hence

$$c_{ij} \dim V_j = \dim V_i \delta_{ij}.$$

If  $j \neq i$ , this gives  $c_{ij} = 0$ . If  $j = i$ , then  $\dim V_j = \dim V_i$ , so  $c_{ii} = 1$ . Thus

$$P_i|_{V_j} = \begin{cases} \text{id}_{V_i}, & j = i, \\ 0, & j \neq i. \end{cases}$$

Therefore  $P_i$  is exactly the projection onto the direct sum of all irreducible summands isomorphic to  $V_i$ . In other words,  $P_i$  is the projection onto the  $V_i$ -isotypic component of  $V$ .

Since it acts as the identity on  $V^{(i)}$  and as zero on its complementary isotypic components, it follows immediately that

$$P_i^2 = P_i.$$

Thus  $P_i$  is indeed a projection operator.

## Application to $\mathbb{Z}_n$ and Fourier modes

Let  $G = \mathbb{Z}_n$ . We write its elements additively as

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}.$$

Let  $V$  be the vector space of functions

$$f : \mathbb{Z}_n \rightarrow \mathbb{C}.$$

This is the regular representation of  $\mathbb{Z}_n$ . The group acts by translation. We choose the convention

$$(\rho(r)f)(m) = f(m - r),$$

where subtraction is modulo  $n$ . This is the same pullback convention used for wave functions: the group element acts on the argument by the inverse translation.

Since  $\mathbb{Z}_n$  is abelian, all irreducible complex representations are one-dimensional. They are indexed by  $k \in \mathbb{Z}_n$  and given by

$$\chi_k(r) = e^{-2\pi i k r / n}.$$

The complex conjugate character is

$$\overline{\chi_k(r)} = e^{2\pi i k r / n}.$$

Since  $\dim V_k = 1$  and  $|\mathbb{Z}_n| = n$ , the projection operator onto the  $k$ -th Fourier mode is

$$P_k = \frac{1}{n} \sum_{r=0}^{n-1} e^{2\pi i k r / n} \rho(r).$$

Applying this to a function  $f : \mathbb{Z}_n \rightarrow \mathbb{C}$ , we obtain

$$\begin{aligned} (P_k f)(m) &= \frac{1}{n} \sum_{r=0}^{n-1} e^{2\pi i k r / n} (\rho(r)f)(m) \\ &= \frac{1}{n} \sum_{r=0}^{n-1} e^{2\pi i k r / n} f(m - r). \end{aligned}$$

Make the change of variables  $s = m - r$ . Then  $r = m - s$ , and hence

$$\begin{aligned} (P_k f)(m) &= \frac{1}{n} \sum_{s=0}^{n-1} e^{2\pi i k (m-s) / n} f(s) \\ &= e^{2\pi i k m / n} \left( \frac{1}{n} \sum_{s=0}^{n-1} f(s) e^{-2\pi i k s / n} \right). \end{aligned}$$

Thus

$$(P_k f)(m) = \hat{f}(k) e^{2\pi i k m / n},$$

where

$$\hat{f}(k) = \frac{1}{n} \sum_{s=0}^{n-1} f(s) e^{-2\pi i k s / n}$$

is the  $k$ -th Fourier coefficient. Therefore  $P_k$  extracts precisely the  $k$ -th Fourier mode.

Summing over all  $k$ , we recover the Fourier decomposition

$$f(m) = \sum_{k=0}^{n-1} \hat{f}(k) e^{2\pi i k m / n}.$$

Thus the isotypic decomposition of the regular representation of  $\mathbb{Z}_n$  is exactly the discrete Fourier decomposition.

## Exercise 2: Dimensions of irreducible representations of $S_n$

We start from the Frobenius formula for the characters of the symmetric group:

$$\chi_\lambda(C_{\vec{i}}) = \left[ \Delta(x) \prod_{j=1}^n P_j(x)^{i_j} \right]_{(l_1, \dots, l_k)}.$$

Here:

- $\lambda = (\lambda_1, \dots, \lambda_k)$  is a partition of  $n$  with  $k$  rows;
- $l_j = \lambda_j + k - j$ ;
- $C_{\vec{i}}$  is the conjugacy class with  $i_j$  cycles of length  $j$ ;
- $P_j(x) = x_1^j + \dots + x_k^j$ ;
- $\Delta(x) = \prod_{p < q} (x_p - x_q)$ ;
- $[f(x)]_{(l_1, \dots, l_k)}$  denotes the coefficient of  $x_1^{l_1} \dots x_k^{l_k}$  in  $f(x)$ .

We want to derive

$$\dim V_\lambda = \frac{n!}{l_1! \dots l_k!} \prod_{i < j} (l_i - l_j).$$

**Solution.** The dimension of a representation is the value of its character at the identity:

$$\dim V_\lambda = \chi_\lambda(e).$$

In  $S_n$ , the identity permutation has cycle type consisting of  $n$  cycles of length 1. Thus the corresponding vector is

$$\vec{i} = (n, 0, 0, \dots, 0).$$

Therefore, in the Frobenius formula, we have

$$\prod_{j=1}^n P_j(x)^{i_j} = P_1(x)^n.$$

Since

$$P_1(x) = x_1 + \dots + x_k,$$

we get

$$\dim V_\lambda = [\Delta(x)(x_1 + \dots + x_k)^n]_{(l_1, \dots, l_k)}.$$

Now write the Vandermonde determinant as

$$\Delta(x) = \prod_{p < q} (x_p - x_q) = \det \left( x_i^{k-j} \right)_{1 \leq i, j \leq k}.$$

Expanding this determinant gives

$$\Delta(x) = \sum_{\sigma \in S_k} \text{sgn}(\sigma) \prod_{i=1}^k x_i^{k-\sigma(i)}.$$

Also, by the multinomial theorem,

$$(x_1 + \dots + x_k)^n = \sum_{r_1 + \dots + r_k = n} \frac{n!}{r_1! \dots r_k!} x_1^{r_1} \dots x_k^{r_k}.$$

We need the coefficient of  $x_1^{l_1} \cdots x_k^{l_k}$  in the product. For a fixed permutation  $\sigma \in S_k$ , the Vandermonde term contributes the power

$$x_i^{k-\sigma(i)}$$

in the variable  $x_i$ . Hence the multinomial term must contribute

$$x_i^{l_i-(k-\sigma(i))} = x_i^{l_i-k+\sigma(i)}.$$

Thus

$$r_i = l_i - k + \sigma(i).$$

The corresponding multinomial coefficient is

$$\frac{n!}{\prod_{i=1}^k (l_i - k + \sigma(i))!}.$$

Therefore

$$\dim V_\lambda = n! \sum_{\sigma \in S_k} \text{sgn}(\sigma) \prod_{i=1}^k \frac{1}{(l_i - k + \sigma(i))!}.$$

This sum is a determinant:

$$\dim V_\lambda = n! \det \left( \frac{1}{(l_i - k + j)!} \right)_{1 \leq i, j \leq k}.$$

It remains to evaluate this determinant. We claim that

$$\det \left( \frac{1}{(l_i - k + j)!} \right)_{1 \leq i, j \leq k} = \frac{\prod_{i < j} (l_i - l_j)}{l_1! \cdots l_k!}.$$

To see this, factor  $1/l_i!$  from row  $i$ . Since

$$\frac{1}{(l_i - k + j)!} = \frac{l_i(l_i - 1) \cdots (l_i - k + j + 1)}{l_i!},$$

we obtain

$$\det \left( \frac{1}{(l_i - k + j)!} \right) = \frac{1}{l_1! \cdots l_k!} \det (l_i(l_i - 1) \cdots (l_i - k + j + 1)).$$

For fixed  $j$ , the entry in column  $j$  is a polynomial in  $l_i$  of degree  $k - j$ . Thus the columns have degrees

$$k - 1, k - 2, \dots, 0.$$

This determinant is an alternating polynomial in the variables  $l_1, \dots, l_k$ . Its highest-degree part is the ordinary Vandermonde determinant

$$\det(l_i^{k-j})_{1 \leq i, j \leq k} = \prod_{i < j} (l_i - l_j).$$

Since both sides are alternating polynomials of the same total degree, they are equal. Hence

$$\det \left( \frac{1}{(l_i - k + j)!} \right) = \frac{\prod_{i < j} (l_i - l_j)}{l_1! \cdots l_k!}.$$

Substituting this into the earlier expression for the dimension gives

$$\dim V_\lambda = \frac{n!}{l_1! \cdots l_k!} \prod_{i < j} (l_i - l_j),$$

as required.

**Remark.** This formula is equivalent to the hook-length formula

$$\dim V_\lambda = \frac{n!}{\prod_{b \in \lambda} h(b)},$$

where  $h(b)$  is the hook length of the box  $b$  in the Young diagram of  $\lambda$ . The formula in this exercise is the version naturally obtained from the Frobenius character formula.

### Exercise 3: Schur functors

Let  $G$  be a group and let  $(V, \rho)$  be a representation of  $G$ . We consider the tensor power

$$V^{\otimes n} = V \otimes \cdots \otimes V.$$

#### (a) The actions of $G$ and $S_n$ commute

The group  $G$  acts on  $V^{\otimes n}$  by the tensor product representation:

$$\rho^{\otimes n}(g)(v_1 \otimes \cdots \otimes v_n) = \rho(g)v_1 \otimes \cdots \otimes \rho(g)v_n.$$

The symmetric group  $S_n$  acts by permuting tensor factors. We use the convention

$$\sigma(v_1 \otimes \cdots \otimes v_n) = v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)}.$$

**Solution.** We compute both possible orders of applying  $g \in G$  and  $\sigma \in S_n$ . First,

$$\begin{aligned} \sigma \rho^{\otimes n}(g)(v_1 \otimes \cdots \otimes v_n) &= \sigma(\rho(g)v_1 \otimes \cdots \otimes \rho(g)v_n) \\ &= \rho(g)v_{\sigma(1)} \otimes \cdots \otimes \rho(g)v_{\sigma(n)}. \end{aligned}$$

On the other hand,

$$\begin{aligned} \rho^{\otimes n}(g)\sigma(v_1 \otimes \cdots \otimes v_n) &= \rho^{\otimes n}(g)(v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)}) \\ &= \rho(g)v_{\sigma(1)} \otimes \cdots \otimes \rho(g)v_{\sigma(n)}. \end{aligned}$$

The two expressions are equal. Hence

$$\sigma \rho^{\otimes n}(g) = \rho^{\otimes n}(g)\sigma$$

for all  $\sigma \in S_n$  and  $g \in G$ .

Therefore  $S_n$  acts by maps of  $G$ -representations, and equivalently  $G$  acts by maps of  $S_n$ -representations.

#### (b) The multiplicity spaces $S^\lambda(V)$ are representations of $G$

Assume that, as an  $S_n$ -representation,  $V^{\otimes n}$  decomposes as

$$V^{\otimes n} \cong \bigoplus_{\lambda} S^\lambda(V) \otimes V_\lambda,$$

where  $V_\lambda$  is the irreducible representation of  $S_n$  indexed by the partition  $\lambda$  of  $n$ .

**Solution.** The important point is that the actions of  $G$  and  $S_n$  commute. Therefore the action of  $G$  preserves the isotypic components for the  $S_n$ -action.

Indeed, fix a partition  $\lambda$ . The  $\lambda$ -isotypic component of  $V^{\otimes n}$  is the sum of all subrepresentations isomorphic to copies of  $V_\lambda$ . This is precisely the summand

$$S^\lambda(V) \otimes V_\lambda.$$

Let  $w$  lie in this isotypic component, and let  $g \in G$ . Since the  $G$ -action commutes with the  $S_n$ -action, the vector  $\rho^{\otimes n}(g)w$  transforms under  $S_n$  in the same isotypic type. Hence

$$\rho^{\otimes n}(g)w \in S^\lambda(V) \otimes V_\lambda.$$

Thus the  $\lambda$ -isotypic component is stable under  $G$ .

Since  $G$  acts trivially on the label  $V_\lambda$  and commutes with the  $S_n$ -action, the nontrivial part of the  $G$ -action is on the multiplicity space  $S^\lambda(V)$ . Hence  $S^\lambda(V)$  is naturally a representation of  $G$ .

**Example 0.1** (The case  $n = 2$ ). The representation of  $S_2$  on  $V \otimes V$  decomposes into its irreducible components:

$$V \otimes V \cong \text{Sym}^2 V \oplus \Lambda^2 V,$$

where

$$\text{Sym}^2 V = \{v \otimes w + w \otimes v\}, \quad \Lambda^2 V = \{v \otimes w - w \otimes v\}.$$

Here  $\text{Sym}^2 V$  carries the trivial representation of  $S_2$ , while  $\Lambda^2 V$  carries the sign representation.

We now show that these subspaces are stable under the action of  $G$ . Let  $g \in G$ .

For a symmetric tensor,

$$w = v \otimes u + u \otimes v,$$

we have

$$\rho^{\otimes 2}(g)(w) = \rho(g)v \otimes \rho(g)u + \rho(g)u \otimes \rho(g)v,$$

which is again symmetric. Hence  $\text{Sym}^2 V$  is  $G$ -invariant.

Similarly, for an antisymmetric tensor,

$$w = v \otimes u - u \otimes v,$$

we obtain

$$\rho^{\otimes 2}(g)(w) = \rho(g)v \otimes \rho(g)u - \rho(g)u \otimes \rho(g)v,$$

which is again antisymmetric. Hence  $\Lambda^2 V$  is also  $G$ -invariant.

Thus both subspaces are preserved by  $G$ , and we obtain a decomposition of  $V \otimes V$  as a representation of  $G \times S_2$ :

$$V \otimes V \cong \text{Sym}^2 V \otimes \text{triv} \oplus \Lambda^2 V \otimes \text{sign}.$$

In particular,  $\text{Sym}^2 V$  and  $\Lambda^2 V$  are representations of  $G$ .

### (c) The Schur functors for $\lambda = (n)$ and $\lambda = (1, \dots, 1)$

**Solution.** The partition  $(n)$  corresponds to a Young diagram with one row of length  $n$ . The corresponding Young symmetrizer only symmetrizes along this row. Therefore its image consists of completely symmetric tensors:

$$S^{(n)}(V) = \text{Sym}^n(V).$$

Thus the Schur functor for the partition  $(n)$  is the  $n$ -th symmetric power.

The partition  $(1, \dots, 1)$  corresponds to a Young diagram with one column of height  $n$ . The corresponding Young symmetrizer antisymmetrizes along this column. Therefore its image consists of completely antisymmetric tensors:

$$S^{(1, \dots, 1)}(V) = \bigwedge^n V.$$

Thus the Schur functor for the partition  $(1, \dots, 1)$  is the  $n$ -th exterior power.

### The case $\lambda = (2, 1)$ for a two-dimensional representation

Now let  $V$  be a two-dimensional representation of  $G$ . We compute  $S^{(2,1)}(V)$  using the Young symmetrizer.

Let  $e_1, e_2$  be a basis of  $V$ . Consider the partition  $(2, 1)$ , whose Young diagram has two boxes in the first row and one box in the second row: a Young diagram of shape  $2, 1$ . If the canonical tableau is

$$\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$$

then the row group is generated by  $(12)$  and the column group is generated by  $(13)$ . Hence

$$a = e + (12), \quad b = e - (13),$$

and the Young symmetrizer is

$$c_{(2,1)} = ab = (e + (12))(e - (13)).$$

With the convention used in the lectures, this can be written as

$$c_{(2,1)} = e + (12) - (13) - (132).$$

The Schur functor is

$$S^{(2,1)}(V) = \text{im} (c_{(2,1)} : V^{\otimes 3} \rightarrow V^{\otimes 3}).$$

Because  $\dim V = 2$ , this representation has a particularly simple description:

$$S^{(2,1)}(V) \cong V \otimes \bigwedge^2 V.$$

Since  $\bigwedge^2 V$  is one-dimensional, this is a two-dimensional representation of  $G$ . If  $G$  acts on  $V$  by matrices  $\rho(g)$ , then it acts on  $\bigwedge^2 V$  by the determinant:

$$g \cdot (e_1 \wedge e_2) = \det(\rho(g))(e_1 \wedge e_2).$$

Therefore

$$S^{(2,1)}(V) \cong V \otimes \det(V).$$

Here  $\det(V)$  denotes the one-dimensional determinant representation  $\bigwedge^2 V$ .

Let us see this concretely. Acting with the Young symmetrizer on tensors with two copies of one basis vector and one copy of the other gives two independent vectors. For example, up to harmless nonzero scalar factors, we may take

$$u_1 = e_1 \otimes e_1 \otimes e_2 - e_2 \otimes e_1 \otimes e_1,$$

and

$$u_2 = e_2 \otimes e_2 \otimes e_1 - e_1 \otimes e_2 \otimes e_2.$$

These span the image of  $c_{(2,1)}$  in  $V^{\otimes 3}$ . Hence

$$\dim S^{(2,1)}(V) = 2.$$

A basis-free way to write the same space is

$$S^{(2,1)}(V) \cong V \otimes \bigwedge^2 V.$$

For example, one may identify

$$e_1 \otimes (e_1 \wedge e_2) \longleftrightarrow u_1, \quad e_2 \otimes (e_1 \wedge e_2) \longleftrightarrow u_2,$$



up to a choice of normalization and convention for the Young symmetrizer.

Thus, for a two-dimensional representation  $V$  of  $G$ , the Schur functor corresponding to the partition  $(2, 1)$  is

$$S^{(2,1)}(V) \cong V \otimes \bigwedge^2 V.$$

Equivalently, if  $\det(V) = \bigwedge^2 V$ , then

$$S^{(2,1)}(V) \cong V \otimes \det(V).$$